Teaching document reference pack

REX_SYNTHETIC_DEMO | Fictional TA example. No real academy offering, instructor, accreditation or tender.

Participant guide

For the Python Foundations Demo learner. Before class install Python and create a practice folder. After each module complete the course exercise, record one difficulty and ask for feedback. Final task: submit a CSV expense tracker with a README and three validation tests.

Facilitator guide

Authored demonstration reference. Session structure: explain for 20 minutes, demonstrate for 20, practice for 90, review for 30 and reflect for 20. Check understanding using a small numeric example before introducing CSV input. Support learners with a worked example, then remove the scaffolding.

Presentation outline

Slide 1: learning goal and fictional context. Slide 2: data types and conversion. Slide 3: rejecting negative values. Slide 4: totaling expenses with a loop. Slide 5: reusable functions. Slide 6: reading CSV safely. Slide 7: expense tracker demonstration. Slide 8: reflection and next steps. This is an outline, not a generated slide deck.

Question bank

Q1: What is sum([120,80,50])? Answer: 250. Q2: Why validate float input? Answer: invalid input must produce a clear error rather than crash. Q3: When should a function be used? Answer: to isolate and reuse a coherent operation. Practical question: implement a total function that rejects negative expenses.

Assessment rubric

Correctness 50 points: totals and CSV round trip work. Validation 30 points: missing, nonnumeric and negative values handled clearly. Clarity 20 points: meaningful names, a short README and repeatable test steps. Passing demonstration threshold: 60 out of 100. This threshold is an authored fixture, not academy policy.

Capstone specification

Build a local expense tracker with add, list and total commands. Use only fictional transactions. Persist a date, category and amount in CSV. Deliver source code, test inputs, expected totals and a README. Exclude authentication, payments and external personal data from this learning exercise.

Course description

Python Foundations Demo is a fictional 24-hour beginner online course at a demo price of 2400 EGP. It teaches variables, conditions, loops, functions and CSV handling. No programming experience is required. No job, certificate or accreditation is promised.

Curriculum example

Authored reference fixture only; the Curriculum Agent is not implemented by this file. Allocate 6 hours each to getting started, control flow, functions and files, and the expense tracker project. Map each module exercise to the corresponding learning outcome and collect the final project for review.

Training proposal

Fictional client: Cedar Demo Company. Need: introduce ten office staff to Python automation. Proposed scope: the 24-hour Python Foundations Demo course, online delivery, exercises and a final tracker. Demo tuition for ten learners before any eligible demo discount: 24000 EGP. Venue, dates and acceptance require human confirmation. This is not a commercial quote.

Learning outcomes and assessment plan

Outcome 1 write conditions and loops: inspect a totaling exercise. Outcome 2 read and validate CSV: test missing and malformed rows. Outcome 3 build a tracker: review the capstone and rubric. Collect formative feedback every session and use the final project for a summative assessment. No external accreditation claim is made.